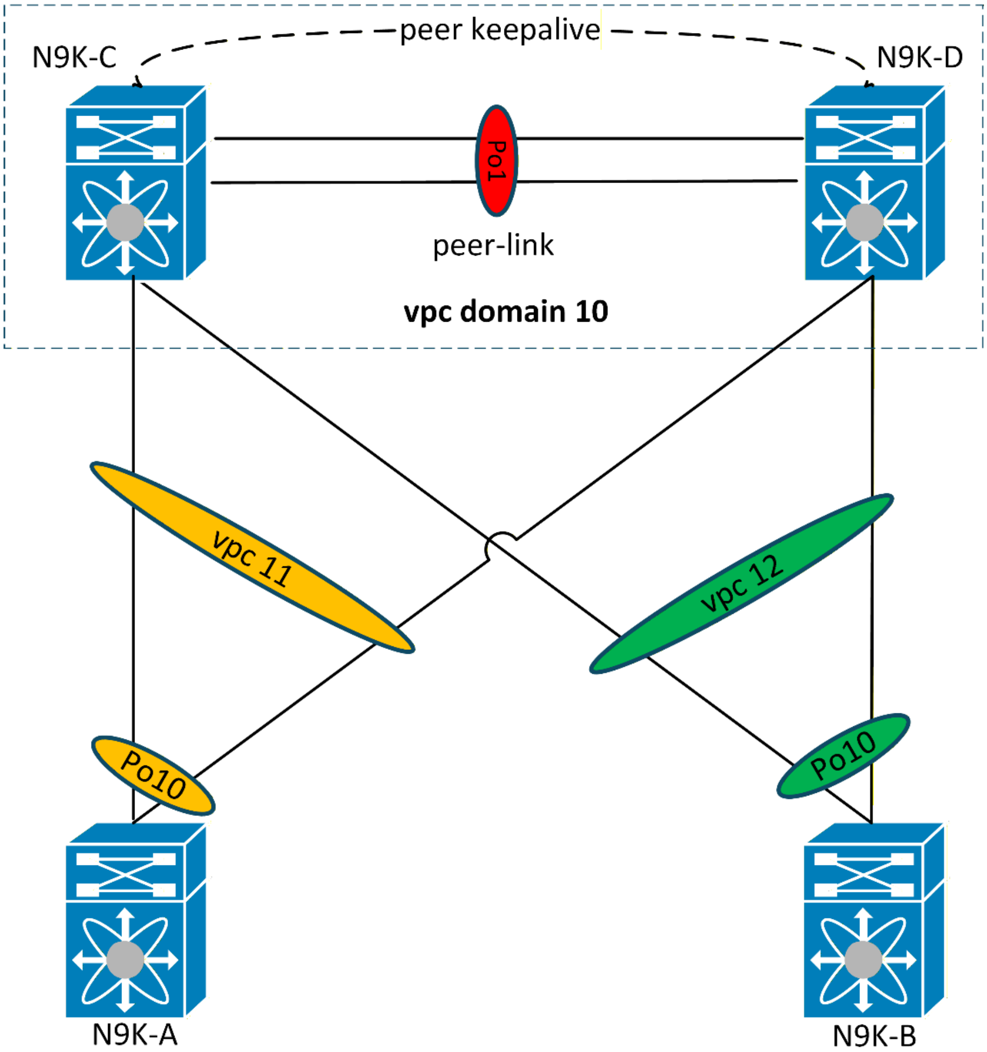
# Cisco virtual Port-Channel (vPC) – Basic Configuration - CLI


**Titus Majeza**
(https://www.linkedin.com/in/titus-majeza/)

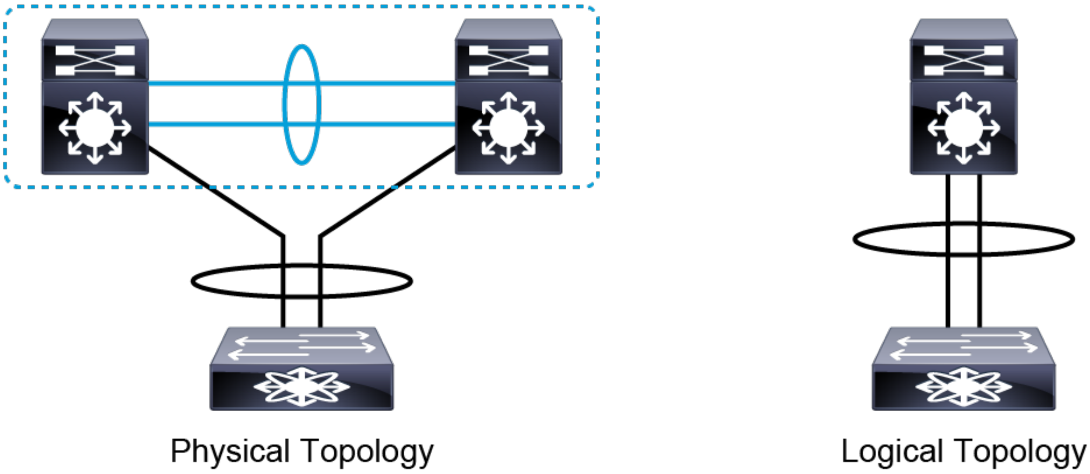

For more labs visit my GitHub repo: https://github.com/TitusM/Cisco-Data-Center

**Note**
This lab was conducted in a controlled environment. Any configurations in a production network should be implemented during a designated maintenance window. Additionally, always refer to official documentation relevant to your specific hardware and software.

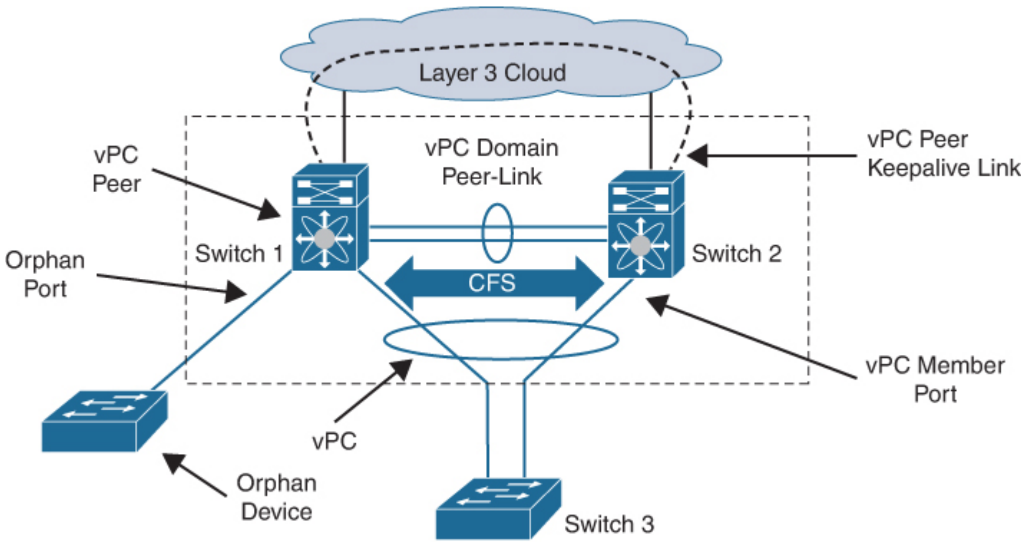
*"It does not matter how slowly you go as long as you do not stop." — Confucius*

# Virtual Port Channel (vPC)

In traditional spanning-tree designs, redundant Layer 2 links between switches are blocked to prevent loops — leaving expensive bandwidth unused and creating suboptimal traffic paths. Virtual Port Channel (vPC) eliminates this limitation by allowing two physical Cisco Nexus switches to appear as a single logical switch to downstream devices. This enables all links to actively forward traffic, effectively doubling the available bandwidth while providing redundancy. If one vPC peer fails, the remaining peer continues to forward traffic with no disruption to the downstream devices.




Before jumping into the configurations, this section will go through a high-level overview of the virtual Port-channel (vPC) architecture and components.




1. **vPC domain**

The vPC domain is formed when two physical switches are linked together and appear as one logical switch. Only two devices can be part of the same vPC domain. The domain includes both vPC peer devices, the vPC peer-keepalive link, the vPC peer link, and all port channels in the vPC domain that are connected to the downstream devices.

2. **vPC peers**

The pair of physical switches in a vPC domain are known as vPC peers. A maximum of two switches can be vPC peers in a vPC domain.

3. **vPC peer-keepalive link**

The peer-keepalive link is a Layer 3 link that often runs over an out-of-band (OOB) network. It provides a communication path that is used as a secondary test to decide whether the remote peer is operating properly. The peer-keepalive status helps the vPC switch determine if the peer-link itself has failed or if the vPC peer has failed entirely. A separate Layer 3 network can also be used for vPC peer-keepalive, often in a dedicated VRF. No data or synchronization traffic is sent over the vPC peer-keepalive link.

4. **vPC peer-link**

The vPC peer-link is used to create the illusion of a single control plane by forwarding traffic between switches in vPC. Control plane information such as spanning-tree BPDUs and LACP packets are exchanged over the peer link. The peer link is also used to synchronize MAC address tables, ARP caches, and IGMP tables that are built via IGMP snooping so that both switches can function in unison.

5. **Cisco Fabric Services over Ethernet (CFSoE)**

The CFSoE is a reliable messaging protocol that is designed to support rapid, stateful, configuration-message passing, and synchronization. Cisco Fabric Services over Ethernet is used to synchronize control plane information between vPC peer switches, to perform vPC configuration and compatibility checks, and to monitor the status of the vPC member ports.

6. **vPC**

This is the combined port-channel that spans both switches in a vPC domain. A vPC is configured towards the downstream devices that support port-channels.

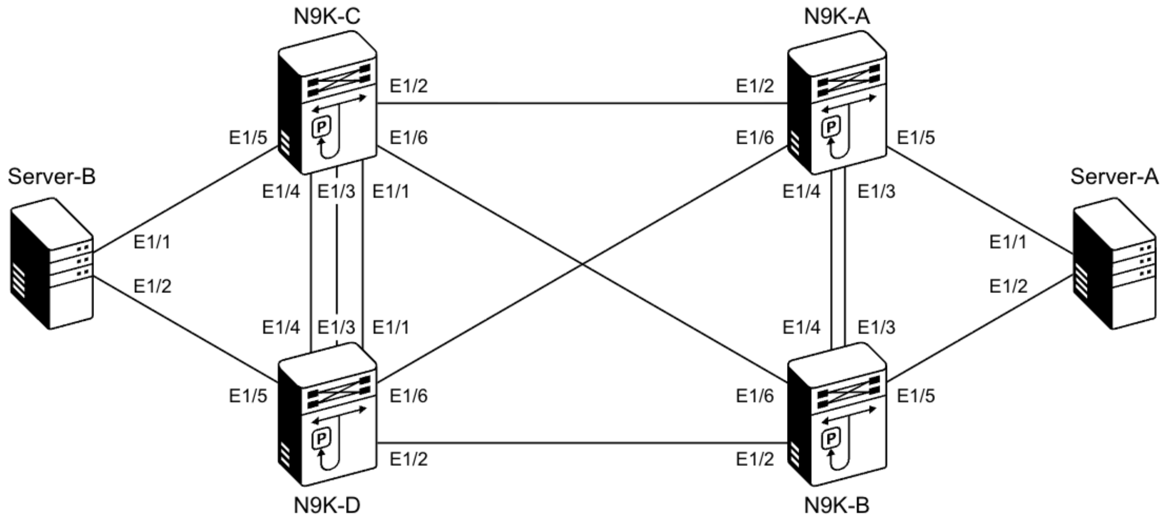
7. **vPC member port**

This is an interface that belongs to a vPC.

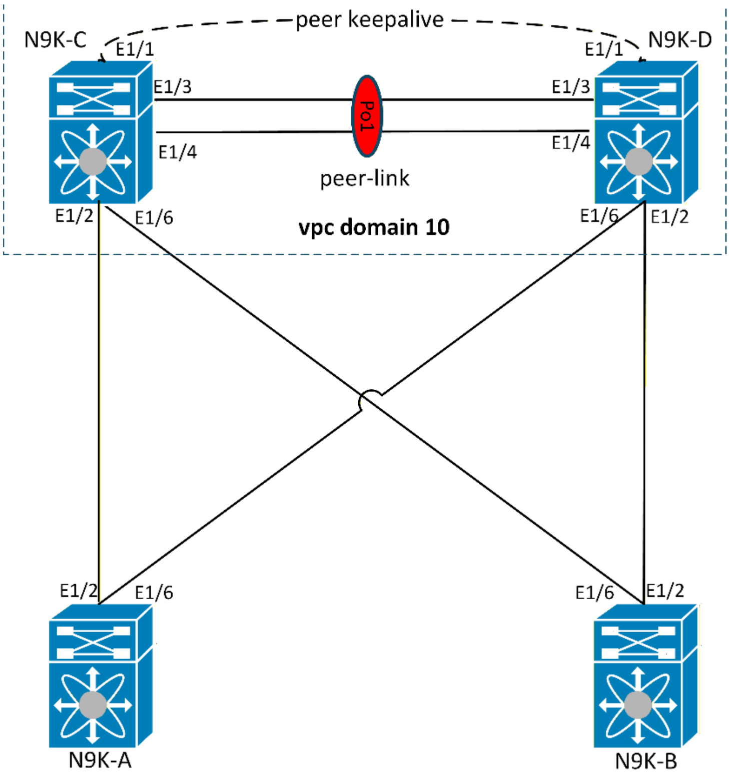
**Some general guidelines for deploying a vPC topology are as follows:**

- A vPC domain has only two peer switches.
- Only one vPC domain can be configured per switch.
- vPC Peer Link must use a 10-Gigabit Ethernet interface at a minimum. Higher-bandwidth interfaces may also be used.
- It is recommended to use at least two dedicated 10-Gigabit Ethernet links for the vPC peer link.
- The downstream device can be any port channel-capable device. The downstream device is not aware that a vPC is present. It only sees a traditional port channel.

## Lab Topology




This lab consists of 4 devices (N9K-A, N9K-B, N9K-C, N9K-D), whereby N9K-C and N9K-D will be configured as vPC peers and N9K-A and N9K-B will be configured as the downstream devices.




## vPC Configurations Steps

No time to read through the step-by-step configuration build up. -> Navigate to the **Full Configurations** section to view the full configurations.

*"It does not matter how slowly you go as long as you do not stop." — Confucius*

### 1. Configure the vPC Domain

Each vPC domain has a vPC instance number that two devices share. The domain ID can be any value between 1 and 1000, and you must configure the same value on both switches that form the vPC pair. **Ensure that the *"vpc"* feature is enabled on the devices.**

| N9K-C | N9K-D |
|---|---|
| feature vpc<br>!<br>vpc domain 10 | feature vpc<br>!<br>vpc domain 10 |

At this point, the vPC is still not operational as the peer-link and peer keep-alive link are not configured as yet.

```
N9K-C# show vpc
Legend:
                (*) - local vPC is down, forwarding via vPC peer-link

vPC domain id                     : 10
Peer status                       : peer link not configured
vPC keep-alive status             : Disabled
Configuration consistency status  : failed
Per-vlan consistency status       : failed
Configuration inconsistency reason: vPC peer-link does not exist
Type-2 consistency status         : failed
Type-2 inconsistency reason       : vPC peer-link does not exist
vPC role                          : none established
Number of vPCs configured         : 0
Peer Gateway                      : Disabled
Dual-active excluded VLANs        : -
Graceful Consistency Check        : Disabled (due to peer
configuration)
Auto-recovery status              : Disabled
Delay-restore status              : Timer is off.(timeout = 30s)
Delay-restore SVI status          : Timer is off.(timeout = 10s)
Delay-restore Orphan-port status  : Timer is off.(timeout = 0s)
Operational Layer3 Peer-router    : Disabled
Virtual-peerlink mode             : Disabled
N9K-C#
```

```
N9K-D# show vpc
Legend:
                (*) - local vPC is down, forwarding via vPC peer-link

vPC domain id                     : 10
Peer status                       : peer link not configured
vPC keep-alive status             : Disabled
Configuration consistency status  : failed
Per-vlan consistency status       : failed
Configuration inconsistency reason: vPC peer-link does not exist
Type-2 consistency status         : failed
Type-2 inconsistency reason       : vPC peer-link does not exist
vPC role                          : none established
Number of vPCs configured         : 0
Peer Gateway                      : Disabled
Dual-active excluded VLANs        : -
Graceful Consistency Check        : Disabled (due to peer
configuration)
Auto-recovery status              : Disabled
Delay-restore status              : Timer is off.(timeout = 30s)
Delay-restore SVI status          : Timer is off.(timeout = 10s)
Delay-restore Orphan-port status  : Timer is off.(timeout = 0s)
Operational Layer3 Peer-router    : Disabled
Virtual-peerlink mode             : Disabled
N9K-D#
```

Notice that the vPC roles and vPC domain MAC address is not yet determined. The vPC peers must first communicate via the vPC peer link to negotiate the vPC role.

```
N9K-C# show vpc role

vPC Role status
----------------------------------------------------
vPC role                        : none established
Dual Active Detection Status    : 0
vPC system-mac                  : 00:00:00:00:00:00
vPC system-priority             : 32667
vPC local system-mac            : 52:92:fa:1f:1b:08
vPC local role-priority         : 0
vPC local config role-priority  : 32667
vPC peer system-mac             : 00:00:00:00:00:00
vPC peer role-priority          : 0
vPC peer config role-priority   : 0
```

```
N9K-D# show vpc role

vPC Role status
----------------------------------------------------
vPC role                        : none established
Dual Active Detection Status    : 0
vPC system-mac                  : 00:00:00:00:00:00
vPC system-priority             : 32667
vPC local system-mac            : 52:90:bf:c0:1b:08
vPC local role-priority         : 0
vPC local config role-priority  : 32667
vPC peer system-mac             : 00:00:00:00:00:00
vPC peer role-priority          : 0
vPC peer config role-priority   : 0
```

### 2. Configure the vPC Peer Keepalive

Configure a vPC peer-keepalive between the N9K-C and N9K-D switches on Ethernet 1/1. Use the IP address 209.165.200.225/24 on N9K-C and 209.165.200.226/24 on N9K-D.

| N9K-C | N9K-D |
|---|---|
| conf t<br>!<br>vrf context VPC-KEEPALIVE<br>!<br>interface ethernet1/1<br>  no switchport<br>  vrf member VPC-KEEPALIVE<br>  ip address 209.165.200.225/24<br>  no shutdown<br>! | conf t<br>!<br>vrf context VPC-KEEPALIVE<br>!<br>interface ethernet1/1<br>  no switchport<br>  vrf member VPC-KEEPALIVE<br>  ip address 209.165.200.226/24<br>  no shutdown<br>! |

Verify that the Layer 3 interfaces are up in the VPC-KEEPALIVE VRF using the `show ip interface brief vrf VPC-KEEPALIVE` command on N9K-C and N9K-D.

```
N9K-C# show ip interface brief vrf VPC-KEEPALIVE
IP Interface Status for VRF "VPC-KEEPALIVE"(4)
Interface            IP Address      Interface Status
Eth1/1               209.165.200.225 protocol-up/link-up/admin-up
```

```
N9K-D# show ip interface brief vrf VPC-KEEPALIVE
IP Interface Status for VRF "VPC-KEEPALIVE"(4)
Interface            IP Address      Interface Status
Eth1/1               209.165.200.226 protocol-up/link-up/admin-up

N9K-D# ping 209.165.200.225 vrf VPC-KEEPALIVE
PING 209.165.200.225 (209.165.200.225): 56 data bytes
36 bytes from 209.165.200.226: Destination Host Unreachable
Request 0 timed out
64 bytes from 209.165.200.225: icmp_seq=1 ttl=254 time=2.504 ms
64 bytes from 209.165.200.225: icmp_seq=2 ttl=254 time=1.529 ms
64 bytes from 209.165.200.225: icmp_seq=3 ttl=254 time=1.374 ms
64 bytes from 209.165.200.225: icmp_seq=4 ttl=254 time=1.799 ms

--- 209.165.200.225 ping statistics ---
5 packets transmitted, 4 packets received, 20.00% packet loss
round-trip min/avg/max = 1.374/1.801/2.504 ms
```

*Note: The first ICMP packet (Request 0) timed out because the ARP table had no entry for the destination. The switch had to broadcast an ARP request and wait for a reply before it could forward the packet. This is normal behavior for the first ping to a new destination.*

Under the vPC domain configure the vPC peer-keepalive link.

```
N9K-C#
vpc domain 10
peer-keepalive destination 209.165.200.226 source 209.165.200.225 vrf VPC-KEEPALIVE
```

```
N9K-D#
vpc domain 10
peer-keepalive destination 209.165.200.225 source 209.165.200.226 vrf VPC-KEEPALIVE
```

At this point, the vPC peers can reach each other via the peer-keepalive link.

| N9K-C# show vpc | N9K-D# show vpc |
|---|---|
| Legend:<br>              (*) - local vPC is down, forwarding via vPC peer-link<br><br>vPC domain id                   : 10<br>Peer status                     : peer link not configured | Legend:<br>              (*) - local vPC is down, forwarding via vPC peer-link<br><br>vPC domain id                   : 10<br>Peer status                     : peer link not configured |

| | |
|---|---|
| vPC keep-alive status : peer is alive | vPC keep-alive status : peer is alive |

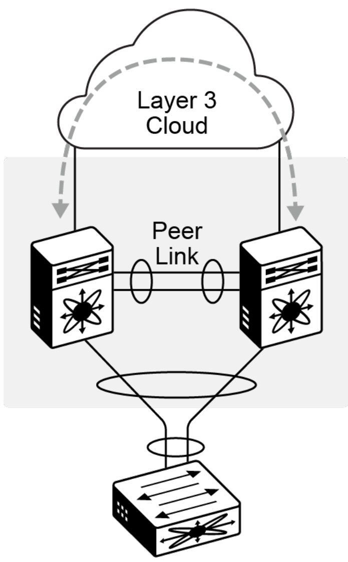
Use the `show vpc peer-keepalive` which shows you various statistics that you can use to verify that the device sends and receives status updates over the intended interface and in the correct VRF instance.

| N9K-C | N9K-D |
|---|---|
| <pre>**N9K-C# show vpc peer-keepalive**<br><br>vPC keep-alive status             : peer is alive<br>--Peer is alive for                : (37) seconds, (577) msec<br>--Send status                      : Success<br>--Last send at                     : 2026.04.06 09:34:44 609 ms<br>--Sent on interface                : Eth1/1<br>--Receive status                   : Success<br>--Last receive at                  : 2026.04.06 09:34:44 722 ms<br>--Received on interface            : Eth1/1<br>--Last update from peer            : (0) seconds, (552) msec<br><br>vPC Keep-alive parameters<br>--Destination                      : 209.165.200.226<br>--Keepalive interval               : 1000 msec<br>--Keepalive timeout                : 5 seconds<br>--Keepalive hold timeout           : 3 seconds<br>--Keepalive vrf                    : VPC-KEEPALIVE<br>--Keepalive udp port               : 3200<br>--Keepalive tos                    : 192</pre> | <pre>**N9K-D# show vpc peer-keepalive**<br><br>vPC keep-alive status             : peer is alive<br>--Peer is alive for                : (97) seconds, (795) msec<br>--Send status                      : Success<br>--Last send at                     : 2026.04.06 09:35:45 470 ms<br>--Sent on interface                : Eth1/1<br>--Receive status                   : Success<br>--Last receive at                  : 2026.04.06 09:35:45 382 ms<br>--Received on interface            : Eth1/1<br>--Last update from peer            : (0) seconds, (713) msec<br><br>vPC Keep-alive parameters<br>--Destination                      : 209.165.200.225<br>--Keepalive interval               : 1000 msec<br>--Keepalive timeout                : 5 seconds<br>--Keepalive hold timeout           : 3 seconds<br>--Keepalive vrf                    : VPC-KEEPALIVE<br>--Keepalive udp port               : 3200<br>--Keepalive tos                    : 192</pre> |

To display various vPC-related statistics, you can use the `show vpc statistics peer-keepalive`

| N9K-C | N9K-D |
|---|---|
| <pre>**N9K-C# show vpc statistics peer-keepalive**<br><br>vPC keep-alive statistics<br>----------------------------------------------------<br>peer-keepalive tx count:          300<br>peer-keepalive rx count:          245<br>average interval for peer rx:     1165<br>Count of peer state changes:      0</pre> | <pre>**N9K-D# show vpc statistics peer-keepalive**<br><br>vPC keep-alive statistics<br>----------------------------------------------------<br>peer-keepalive tx count:          274<br>peer-keepalive rx count:          271<br>average interval for peer rx:     998<br>Count of peer state changes:      0</pre> |

### 3. Configure the vPC Peer-Link



The vPC peer link carries essential vPC traffic between the vPC peer switches. The vPC peer link is a port channel and should consist of at least two dedicated 10 Gigabit Ethernet links.

```
N9K-C
conf t
!
feature lacp
!
interface ethernet1/3-4
  switchport
  switchport mode trunk
  channel-group 1 mode active
  no shut
!
interface port-channel 1
  vpc peer-link
Warning: Bridge Assurance MUST be enabled at the remotely
connected interface
```

```
N9K-D
conf t
!
feature lacp
!
interface ethernet1/3-4
  switchport
  switchport mode trunk
  channel-group 1 mode active
  no shut
!
interface port-channel 1
  vpc peer-link
Warning: Bridge Assurance MUST be enabled at the remotely
connected interface
```

Verify that Bridge-Assurance is enabled on N9K-C and N9K-D.

```
N9K-C# show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: none
L2 Gateway STP                          is disabled
Port Type Default                       is disable
Edge Port [PortFast] BPDU Guard Default  is disabled
Edge Port [PortFast] BPDU Filter Default is disabled
Bridge Assurance                        is enabled
Loopguard Default                       is disabled
Pathcost method used                    is short
STP-Lite                                is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      0         0        0          1          1
VLAN0200                      1         0        0          3          4
---------------------- -------- --------- -------- ---------- ----------
2 vlans                       1         0        0          4          5
```

```
N9K-D# show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0001
L2 Gateway STP                          is disabled
Port Type Default                       is disable
Edge Port [PortFast] BPDU Guard Default  is disabled
Edge Port [PortFast] BPDU Filter Default is disabled
Bridge Assurance                        is enabled
Loopguard Default                       is disabled
Pathcost method used                    is short
STP-Lite                                is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      0         0        0          1          1
VLAN0200                      0         0        0          4          4
---------------------- -------- --------- -------- ---------- ----------
2 vlans                       0         0        0          5          5
```

Verify that the port channel interface (peer-link) is operational.

```
N9K-C# show port-channel summary
Flags:  D - Down         P - Up in port-channel (members)
        I - Individual   H - Hot-standby (LACP only)
        s - Suspended    r - Module-removed
        b - BFD Session Wait
        S - Switched     R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------
1     Po1(SU)     Eth      LACP      Eth1/3(P)    Eth1/4(P)
```

```
N9K-D# show port-channel summary
Flags:  D - Down         P - Up in port-channel (members)
        I - Individual   H - Hot-standby (LACP only)
        s - Suspended    r - Module-removed
        b - BFD Session Wait
        S - Switched     R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------
1     Po1(SU)     Eth      LACP      Eth1/3(P)    Eth1/4(P)
```

The "`show vpc`" status now shows that peer adjacency is correctly formed and and configuration consistency checks have passed.

```
N9K-C# show vpc
Legend:
                (*) - local vPC is down, forwarding via vPC peer-link
```

```
N9K-D# show vpc
Legend:
                (*) - local vPC is down, forwarding via vPC peer-link
```

*"It does not matter how slowly you go as long as you do not stop." — Confucius*

```
vPC domain id                       : 10
Peer status                         : peer adjacency formed ok
vPC keep-alive status               : peer is alive
Configuration consistency status    : success
Per-vlan consistency status         : success
Type-2 consistency status           : success
vPC role                            : secondary
Number of vPCs configured           : 0
Peer Gateway                        : Disabled
Dual-active excluded VLANs          : -
Graceful Consistency Check          : Enabled
Auto-recovery status                : Disabled
Delay-restore status                : Timer is off.(timeout = 30s)
Delay-restore SVI status            : Timer is off.(timeout = 10s)
Delay-restore Orphan-port status    : Timer is off.(timeout = 0s)
Operational Layer3 Peer-router      : Disabled
Virtual-peerlink mode               : Disabled

vPC Peer-link status
---------------------------------------------------------------------
id     Port   Status Active vlans
--     ----   ------ -------------------------------------------------
1      Po1    up     1,200
```

```
vPC domain id                       : 10
Peer status                         : peer adjacency formed ok
vPC keep-alive status               : peer is alive
Configuration consistency status    : success
Per-vlan consistency status         : success
Type-2 consistency status           : success
vPC role                            : primary
Number of vPCs configured           : 0
Peer Gateway                        : Disabled
Dual-active excluded VLANs          : -
Graceful Consistency Check          : Enabled
Auto-recovery status                : Disabled
Delay-restore status                : Timer is off.(timeout = 30s)
Delay-restore SVI status            : Timer is off.(timeout = 10s)
Delay-restore Orphan-port status    : Timer is off.(timeout = 0s)
Operational Layer3 Peer-router      : Disabled
Virtual-peerlink mode               : Disabled

vPC Peer-link status
---------------------------------------------------------------------
id     Port   Status Active vlans
--     ----   ------ -------------------------------------------------
1      Po1    up     1,200
```

The vPC roles have been successfully formed. By default, the switch with the lowest local system-mac is elected as the primary switch in the vPC domain.

```
N9K-C# show vpc role

vPC Role status
----------------------------------------------------
vPC role                         : secondary
Dual Active Detection Status     : 0
vPC system-mac                   : 00:23:04:ee:be:0a
vPC system-priority              : 32667
vPC local system-mac             : 52:92:fa:1f:1b:08
vPC local role-priority          : 32667
vPC local config role-priority   : 32667
vPC peer system-mac              : 52:90:bf:c0:1b:08
vPC peer role-priority           : 32667
vPC peer config role-priority    : 32667
```

```
N9K-D# show vpc role

vPC Role status
----------------------------------------------------
vPC role                         : primary
Dual Active Detection Status     : 0
vPC system-mac                   : 00:23:04:ee:be:0a
vPC system-priority              : 32667
vPC local system-mac             : 52:90:bf:c0:1b:08
vPC local role-priority          : 32667
vPC local config role-priority   : 32667
vPC peer system-mac              : 52:92:fa:1f:1b:08
vPC peer role-priority           : 32667
vPC peer config role-priority    : 32667
```

The vPC peer devices use the vPC domain ID to automatically assign a unique vPC system MAC address. Each vPC domain has a unique MAC address that is a unique identifier for the specific vPC-related operations. This vPC system-mac, is what is presented to downstream devices during LACP exchanges; hence the downstream devices see the devices in a vPC domain as a single device.

Use the `show vpc consistency-parameters global` command to check for a potential vPC configuration inconsistency.

```
N9K-C# show vpc consistency-parameters global

    Legend:
        Type 1 : vPC will be suspended in case of mismatch

Name                        Type  Local Value             Peer Value
-------------               ----  ----------------------  -----------------------
STP MST Simulate PVST       1     Enabled                 Enabled
STP Port Type, Edge         1     Normal, Disabled,       Normal, Disabled,
BPDUFilter, Edge BPDUGuard        Disabled                Disabled
STP MST Region Name         1     ""                      ""
```

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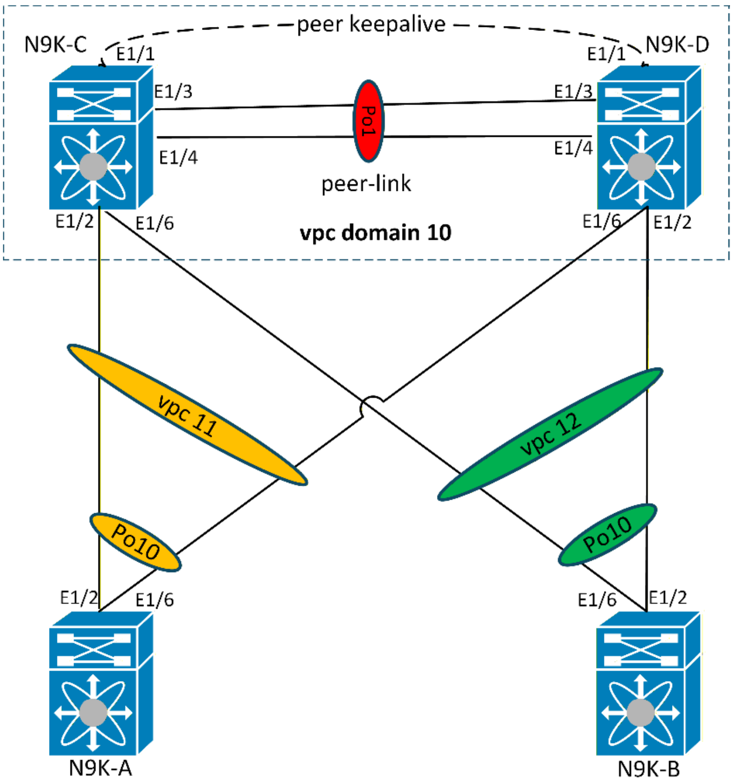
| | | | |
|---|---|---|---|
| STP Disabled | 1 | None | None |
| STP Mode | 1 | Rapid-PVST | Rapid-PVST |
| STP Bridge Assurance | 1 | Enabled | Enabled |
| STP Loopguard | 1 | Disabled | Disabled |
| STP MST Region Instance to VLAN Mapping | 1 | | |
| STP MST Region Revision | 1 | 0 | 0 |
| Interface-vlan admin up | 2 | 200 | 200 |
| Interface-vlan routing capability | 2 | 1,200 | 1,200 |
| Xconnect Vlans | 1 | | |
| QoS (Cos) | 2 | ([0-7], [], [], [], [], [], [], []) | ([0-7], [], [], [], [], [], [], []) |
| Network QoS (MTU) | 2 | (1500, 1500, 1500, 1500, 0, 0, 0, 0) | (1500, 1500, 1500, 1500, 0, 0, 0, 0) |
| Network Qos (Pause: T->Enabled, F->Disabled) | 2 | (F, F, F, F, F, F, F, F) | (F, F, F, F, F, F, F, F) |
| Input Queuing (Bandwidth) | 2 | (0, 0, 0, 0, 0, 0, 0, 0) | (0, 0, 0, 0, 0, 0, 0, 0) |
| Input Queuing (Absolute Priority: T->Enabled, F->Disabled) | 2 | (F, F, F, F, F, F, F, F) | (F, F, F, F, F, F, F, F) |
| Output Queuing (Bandwidth Remaining) | 2 | (100, 0, 0, 0, 0, 0, 0, 0) | (100, 0, 0, 0, 0, 0, 0, 0) |
| Output Queuing (Absolute Priority: T->Enabled, F->Disabled) | 2 | (F, F, F, T, F, F, F, F) | (F, F, F, T, F, F, F, F) |
| Allowed VLANs | - | 1,200 | 1,200 |
| Local suspended VLANs | - | - | - |

## 4. Configure the vPC Member Interfaces




Configure a port channel on the **N9K-C** and **N9K-D** switches to each link toward the **N9K-A** . Use port-channel ID **11**.

```
N9K-C#
!
interface ethernet1/2
  switchport
  switchport mode trunk
  channel-group 11 mode active
  no shutdown
!
interface port-channel 11
  vpc 11
```

```
N9K-D#
!
interface ethernet1/6
  switchport
  switchport mode trunk
  channel-group 11 mode active
  no shutdown
!
interface port-channel 11
  vpc 11
```

Configure a port channel on the **N9K-C** and **N9K-D** switches to each link toward the **N9K-B** switch. Use port-channel ID **12**.

```
N9K-C#
!
interface ethernet1/6
  switchport
  switchport mode trunk
  channel-group 12 mode active
  no shutdown
!
interface port-channel 12
  vpc 12
```

```
N9K-D#
!
interface ethernet1/2
  switchport
  switchport mode trunk
  channel-group 12 mode active
  no shutdown
!
interface port-channel 12
  vpc 12
```

Bind the **N9K-A** and **N9K-B** Ethernet 1/2 and 1/6 interfaces into a port-channel by adding them to the port-channel group 10 using active mode of LACP.

```
N9K-A#
feature lacp
!
interface ethernet1/2,ethernet1/6
  switchport
  switchport mode trunk
  channel-group 10 mode active
  no shutdown
```

```
N9K-B#
feature lacp
!
interface ethernet1/2,ethernet1/6
  switchport
  switchport mode trunk
  channel-group 10 mode active
  no shutdown
```

Verify the port-channel status on all devices.

```
N9K-C# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------
1     Po1(SU)     Eth      LACP      Eth1/3(P)    Eth1/4(P)
11    Po11(SU)    Eth      LACP      Eth1/2(P)
12    Po12(SU)    Eth      LACP      Eth1/6(P)
```

```
N9K-D# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------
1     Po1(SU)     Eth      LACP      Eth1/3(P)    Eth1/4(P)
11    Po11(SU)    Eth      LACP      Eth1/6(P)
12    Po12(SU)    Eth      LACP      Eth1/2(P)
```

```
N9K-A# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
```

```
N9K-B# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
```

| N9K-C | N9K-D |
|---|---|
| s - Suspended r - Module-removed<br>b - BFD Session Wait<br>S - Switched R - Routed<br>U - Up (port-channel)<br>p - Up in delay-lacp mode (member)<br>M - Not in use. Min-links not met<br>--------------------------------------------------------------------------------<br>Group Port- Type Protocol Member Ports<br>Channel<br>--------------------------------------------------------------------------------<br>**10 Po10(SU) Eth LACP Eth1/2(P) Eth1/6(P)** | s - Suspended r - Module-removed<br>b - BFD Session Wait<br>S - Switched R - Routed<br>U - Up (port-channel)<br>p - Up in delay-lacp mode (member)<br>M - Not in use. Min-links not met<br>--------------------------------------------------------------------------------<br>Group Port- Type Protocol Member Ports<br>Channel<br>--------------------------------------------------------------------------------<br>**10 Po10(SU) Eth LACP Eth1/2(P) Eth1/6(P)** |

Verify the vPC status.

| N9K-C | N9K-D |
|---|---|
| **N9K-C# show vpc**<br><br>vPC status<br>--------------------------------------------------------------------------------<br>Id Port Status Consistency Reason Active vlans<br>-- ------------ ------ ----------- ------ --------<br>**11 Po11 up success success 1,200**<br><br>**12 Po12 up success success 1,200**<br><br>Please check "show vpc consistency-parameters vpc <vpc-num>" for the consistency reason of down vpc and for type-2 consistency reasons for any vpc. | **N9K-D# show vpc**<br><br>vPC status<br>--------------------------------------------------------------------------------<br>Id Port Status Consistency Reason Active vlans<br>-- ------------ ------ ----------- ------ --------<br>**11 Po11 up success success 1,200**<br><br>**12 Po12 up success success 1,200**<br><br>Please check "show vpc consistency-parameters vpc <vpc-num>" for the consistency reason of down vpc and for type-2 consistency reasons for any vpc. |

# Full Configurations

| N9K-C | N9K-D |
|---|---|
| feature vpc<br>feature lacp<br>!<br>vrf context VPC-KEEPALIVE<br>!<br>interface ethernet1/1<br>no switchport<br>vrf member VPC-KEEPALIVE<br>ip address 209.165.200.225/24<br>no shutdown<br>!<br>vpc domain 10<br>peer-keepalive destination 209.165.200.226 source 209.165.200.225 vrf VPC-KEEPALIVE<br>!<br>interface ethernet1/3-4<br>switchport<br>switchport mode trunk<br>channel-group 1 mode active<br>no shut<br>!<br>interface port-channel 1<br>vpc peer-link<br>!<br>interface ethernet1/2<br>switchport<br>switchport mode trunk<br>channel-group 11 mode active | feature vpc<br>feature lacp<br>!<br>vrf context VPC-KEEPALIVE<br>!<br>interface ethernet1/1<br>no switchport<br>vrf member VPC-KEEPALIVE<br>ip address 209.165.200.226/24<br>no shutdown<br>!<br>vpc domain 10<br>peer-keepalive destination 209.165.200.225 source 209.165.200.226 vrf VPC-KEEPALIVE<br>!<br>interface ethernet1/3-4<br>switchport<br>switchport mode trunk<br>channel-group 1 mode active<br>no shut<br>!<br>interface port-channel 1<br>vpc peer-link<br>!<br>interface ethernet1/6<br>switchport<br>switchport mode trunk<br>channel-group 11 mode active |

| | |
|---|---|
| <pre>  no shutdown<br>!<br>interface port-channel 11<br>  vpc 11<br>!<br>interface ethernet1/6<br>  switchport<br>  switchport mode trunk<br>  channel-group 12 mode active<br>  no shutdown<br>!<br>interface port-channel 12<br>  vpc 12</pre> | <pre>  no shutdown<br>!<br>interface port-channel 11<br>  vpc 11<br>!<br>interface ethernet1/2<br>  switchport<br>  switchport mode trunk<br>  channel-group 12 mode active<br>  no shutdown<br>!<br>interface port-channel 12<br>  vpc 12</pre> |

| **N9K-A#** | **N9K-B#** |
|---|---|
| <pre>feature lacp<br>!<br>interface ethernet1/2,ethernet1/6<br>  switchport<br>  switchport mode trunk<br>  channel-group 10 mode active<br>  no shutdown</pre> | <pre>feature lacp<br>!<br>interface ethernet1/2,ethernet1/6<br>  switchport<br>  switchport mode trunk<br>  channel-group 10 mode active<br>  no shutdown</pre> |

For more labs visit my GitHub repo:
https://github.com/TitusM/Cisco-Data-Center

# References


**Cisco U courses:**

1. Understanding Cisco Data Center Foundations | DCFNDU
2. Troubleshooting Cisco Data Center Infrastructure | DCIT
3. Designing Cisco Data Center Infrastructure | DCID

**Cisco Official Whitepaper(s):**

1. https://www.cisco.com/c/en/us/td/docs/dcn/nx-os/nexus9000/105x/configuration/interfaces/cisco-nexus-9000-series-nx-os-interfaces-configuration-guide-release-105x/m_configuring_vpcs_9x.html#_0524b521-5191-4eab-8354-6942d82737f8